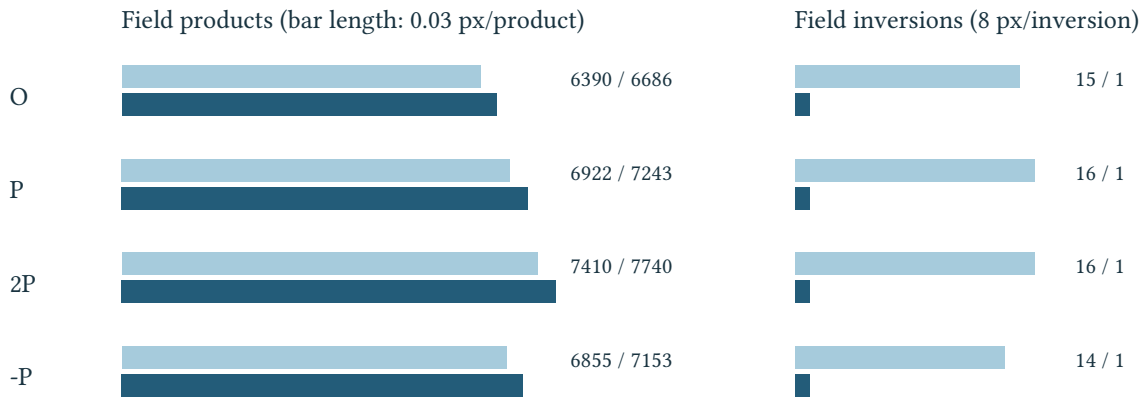
Joint conversion: shifted points and source lanes

2026-11-03 · Exact geometry arithmetic · Four fixed public fixtures

The native entry point now takes only Q and a prefix length. It generates shifted points projectively and converts them together with finite grid lanes. Prepared strides remain reusable. No scalar labels or precomputed shifted points are inputs.



Light: original receipt | Dark: joint native | Four fixed fixtures; exact counts, no timings

Checked tradeoff. On O,P,2P,-P, target geometry inversions fall from15/16/16/14 to1 each. Field products increase by296/321/330/298. Counts include original prefix generation versus projective generation, shared conversion and new point validation; setup is excluded from both target tallies. Products include squares and constant multiplies. Additions, native integer work and memory costs remain unpriced, so this chart does not establish elapsed-time or complete-cost savings.

Derivation. Co-converting K finite shifted points adds7K products to the existing batch, with no extra inversion. K is15/16/16/15 here. Separate affine prefix inversions disappear. Projective prefix products and exceptional cases were derived before the test and included in its assertions.

Native checks. At source20ab3b2a9083, one rustc1.99.1 build and one unit invocation passed3 methods. All64 generated shifted-point entries and1426 finite golden slots agree, including exact projective coordinates and inverse-Z values. Identity/empty/invalid/singleton controls pass. Product/inversion hooks are cfg(test) only; the production prefix otherwise matches the preserved window.

Scope and next. This is fixed-width geometry over the original F509 fixture; prime/subgroup premises remain caller-supplied. No sign oracle, resultant, scalar-recovery cohort or cloud job ran. Connect normalized recurrence and the original product-tree/multipoint resultant before admitting the complete native 4,563-case cohort. Larger-field backend and full costs remain explicit tasks.

Evidence. check-receipt.txt, test.log, counts.json, original baseline TSV, source/input hashes and golden fixtures. DERIVATION.md gives exact exclusions. The chart contains four exact software/count fixtures, no timing sample or uncertainty estimate. Original formula/source references are retained there.